Course Code: 2165119 **Course Name:** ASD&D Lab

Class: TE **Batch:** T23 **Semester:** V

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EXPERIMENT NO. 06

Aim: - To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Theory:

Agile Software Development is an iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end. It emphasizes flexibility, continuous improvement, and rapid response to change.

Scrum is one of the most popular Agile frameworks. It breaks work down into smaller, manageable chunks called "Sprints" (usually 1-4 weeks long) where a cross-functional team works to complete a specific set of tasks.

Jira is a powerful issue and project tracking tool developed by Atlassian, widely used by Agile teams.

- **Epic:** A large body of work that can be broken down into smaller tasks or user stories.
- **Backlog:** A prioritized list of work for the development team that is derived from the roadmap and its requirements.
- **Sprint Board:** A visual representation (often Kanban-style) of the tasks in the current sprint, moving through stages like "To Do", "In Progress", and "Done".

DevOps Integration Context: In a modern CI/CD environment, Jira can be integrated with GitHub and Jenkins. Developers can include the Jira Issue Key (e.g., SCRUM-11) in their Git commit messages. This automatically links the code changes to the Jira ticket and can even transition the issue automatically from "In Progress" to "Done" when the pipeline successfully deploys the code.

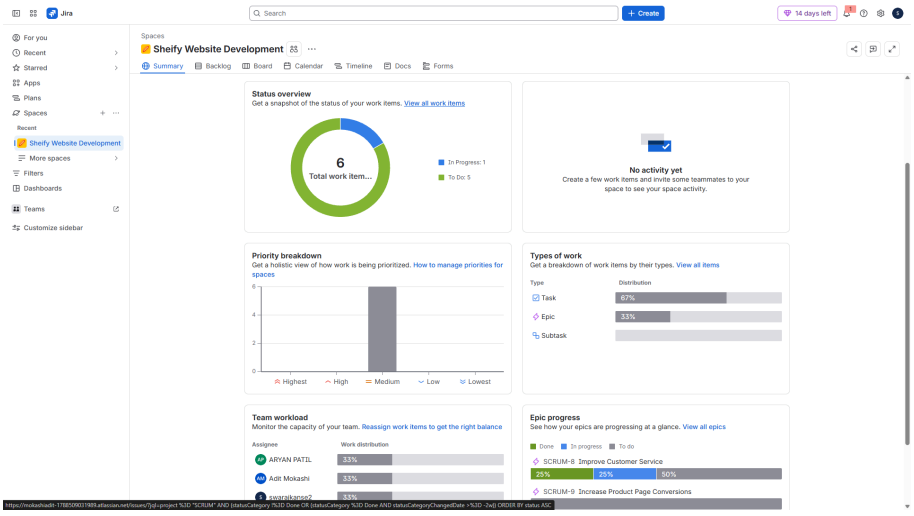
IMPLEMENTATION

Step 1: Project Initialization and Epic Creation

A new Scrum software development project named **"Sheify Website Development"** was created in Jira. High-level goals were defined as Epics, such as "Improve Customer Service" and "Increase Product Page Conversions".

Step 2: Backlog Grooming and Sprint Planning

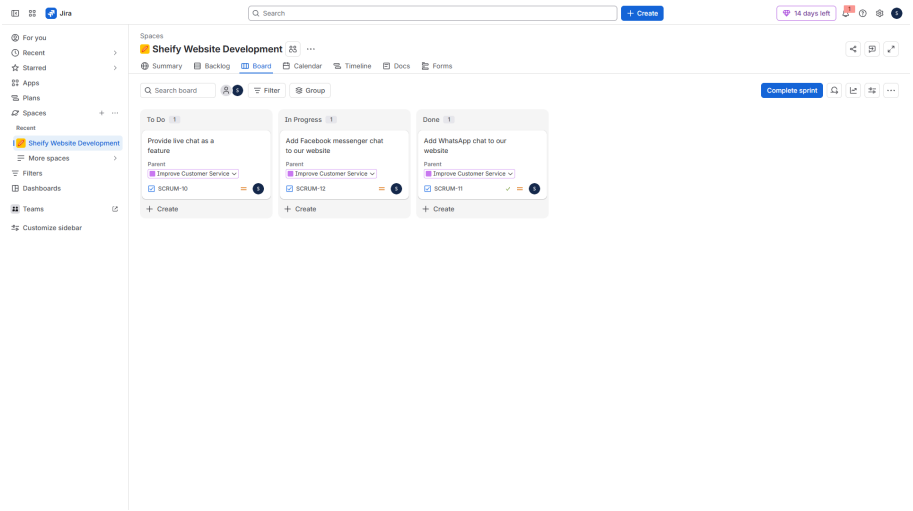
Tasks and user stories were added to the overall Project Backlog. For the first iteration, specific tasks related to the "Improve Customer Service" Epic (e.g., adding WhatsApp, Facebook Messenger, and Live Chat features) were selected and moved into **Scrum Sprint 1**. Tasks were assigned to team members (Swaraj Kanse, Aryan Patil, Adit Mokashi) and prioritized.



Step 3: Executing the Active Sprint

Once the Sprint was started, the team used the Agile Scrum Board to track daily progress. The board visually categorized work into three columns: **To Do**, **In Progress**, and **Done**.

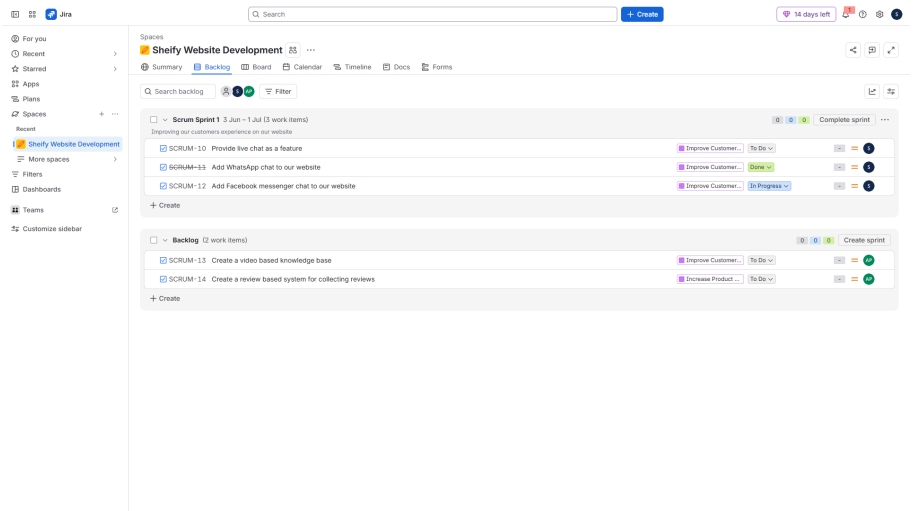
For example, the task "Add WhatsApp chat to our website" (SCRUM-11) was completed and moved to the "Done" column, while the Facebook messenger task was moved to "In Progress".



Step 4: Project Monitoring and Reporting

The project's overall health and velocity were monitored using the Jira Summary Dashboard. This dashboard provided real-time analytics, including:

- **Status Overview:** A visual donut chart showing the ratio of To Do, In Progress, and Done items.
- **Team Workload:** The distribution of tasks among team members to prevent bottlenecks.
- **Epic Progress:** A progress bar showing the completion percentage of high-level goals like the "Improve Customer Service" epic (currently at 25% Done, 25% In Progress, 50% To Do).



Conclusion: Thus, we have successfully studied the Agile Software Project lifecycle and utilized Jira to create a project, manage a product backlog, plan a Sprint, visually track task progression via a Scrum board, and monitor overall team performance using dashboard analytics.